PPP tube power amplifier, 4x KT88 parallel push–pull (circlotron), mono block

Amplifier

Datei: amp.kicad_sch

Power Supply

Datei: supply.kicad_sch

Sheet: /
File: ppp.kicad_sch

Title: PPP power amplifier 4x KT88 (mono)

Size: A4	Date: 2026-10-20	**Rev: 1**
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PPP 4x KT88 – Amplifier

J1 Input a/b/M
T1 E-1220
C1 2.2µ/40V bip.
R1 68k
R2 68k
R3 2.2k
C2 10p
C3 100p
V1A ECC83
R4 150k
R5 2.7k
C4 47µ/450V
D1 56V
D2 100V
D3 100V
D4 100V
C5 47µ/450V
R6 2.2k
C7 0.22µ/630V
V1B ECC83
R7 1M
R8 2.7k
R9 47k
R10 220k
C6 3.3p
R11 47k
R12 47k
R13 2.7k
R14 33k
R15 10k
R16 10k
V2A ECC81
V2B ECC81
C10 22µ/100V
C8 0.22µ/630V
C11 0.22µ/630V
R31 10k
R33 10k
V3A KT88
V5A KT88
R43 150R
R45 150R
R19 150k
R21 150k
R37 10R
R39 10R
R25 33k
R27 33k
P1 25k
P3 25k
C17 47µ/100V
C15 47µ/100V
R17 6.8k
D5 1N4007
R18 6.8k
D6 1N4007
C18 47µ/100V
C16 47µ/100V
P2 25k
P4 25k
R26 33k
R28 33k
R38 10R
R40 10R
R20 150k
R22 150k
R32 10k
R34 10k
V4A KT88
V6A KT88
C9 0.22µ/630V
C12 0.22µ/630V
R44 150R
R46 150R
T2 AP-634/2
Rkk 330Ω, >140W
J3 Speaker 8/6/4/0
G1
A1
M1
UG1
UG2
M2
A2
G2
GND

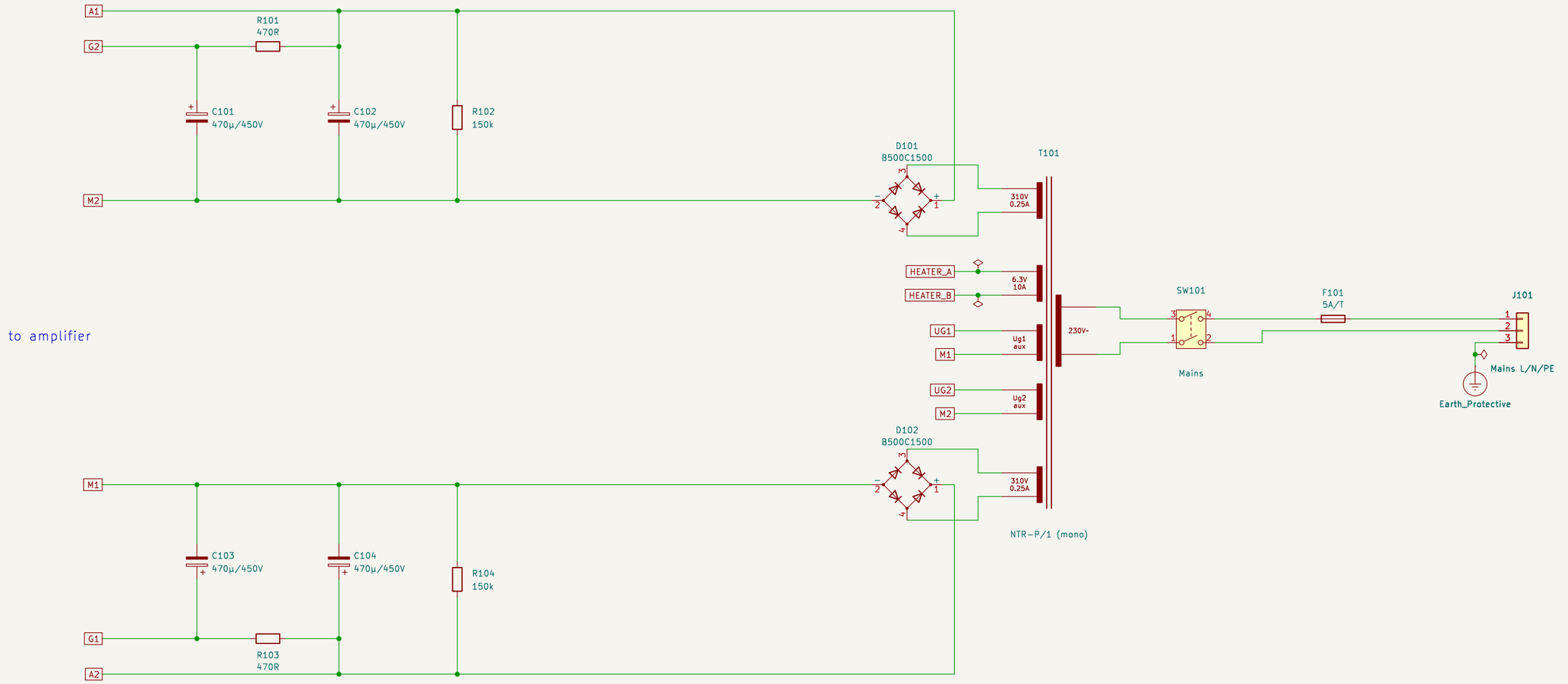
6.3V~ heaters (from mains transformer), hum balance R49/R50

V1C ECC83
V2C ECC81
V3B KT88
V4B KT88
V5B KT88
V6B KT88
R49 47R
R50 47R
HEATER_A
HEATER_B

Original designators: V1=R81 (ECC83), V2=R82 (ECC81), V3..V6=R83..R86 (KT88 variant, original EL34; R87/R88 omitted). T1=Ü1 (E-1220). T2=Ü2 (AP-634/2).
Supply rails A1/G1/M1, A2/G2/M2 and UG1/UG2 come from the power supply sheet.
Upper tubes: anode A1, screen G1, cathode M1 (G1 is referenced to M1).
Lower tubes: anode A2, screen G2, cathode M2 (G2 is referenced to M2).
Ug1/Ug2: bias windings, floating on M1/M2; bias adjust P1..P4.

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PPP 4x KT88 – Power supply



to amplifier

Mono half of the original stereo supply (NTR-P/1: 4x 310V, 2x 6.3V).
Aux windings Ug1/Ug2 added for the grid bias supplies (approx. 60V~ / 50mA, value TBD).
UG1 winding floats on M1, UG2 winding floats on M2.

Sheet: /Power Supply/		
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Title: PPP power amplifier 4x KT88 – power supply (mono)		
Size: A3	Date: 2026-10-20	**Rev: 1**
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